Mathematical and Statistical Approach to Defining Meal Boundaries in NR Phase of Order 1 Mice

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1 Introduction

Understanding the feeding behavior of mice is pivotal in studies related to metabolism, neurobiology, and behavioral science. Accurately defining meal boundaries allows researchers to quantify feeding patterns, assess satiation levels, and explore the underlying biological mechanisms governing eating behavior.

This documentation outlines the **mathematical and statistical methods** employed to analyze pellet consumption data, culminating in the establishment of a threshold to delineate meal boundaries effectively. The focus is on the **NR (No-Restriction) phase of Order 1 mice**, encompassing both male and female subjects due to their equivalent pellet intake patterns.

2 Data Collection and Preparation

2.1 Dataset Overview

The dataset comprises pellet consumption data for Order 1 mice during the NR phase. The key variables are:

- **mouse_id**: Unique identifier for each mouse.
- **order**: Order number of the experiment (focused on Order 1).
- **sex**: Gender of the mouse (Male - M, Female - F).
- **diet_phase**: Phase of the diet (NR - No-Restriction).
- **a**: Parameter P_0 representing the initial probability of adding a second pellet.
- **b**: Parameter λ representing the decay rate in the exponential decay model.

2.2 Filtered Data for NR Phase of Order 1 Mice

To maintain focus and ensure consistency in analysis, only data corresponding to the NR phase and Order 1 mice are considered. Both male and female mice are included due to their comparable pellet intake.

Table 1: NR Phase Data for Order 1 Mice

mouse_id	**order**	**sex**	**diet_phase**	**a**	**b**
FEDXA01	2	M	NR	84.8257	0.2172
FEDXA02	2	M	NR	167.0623	0.3675
FEDXA03	2	M	NR	113.2411	0.2469
FEDXA04	2	M	NR	103.8889	0.2542
FEDXA05	2	M	NR	196.9724	0.3446
FEDXA06	2	M	NR	231.2507	0.4319
FEDXA07	1	M	NR	128.9930	0.3354
FEDXA08	1	M	NR	103.8890	0.2542
FEDXA09	1	M	NR	163.1028	0.3759
FEDXA10	1	M	NR	192.0717	0.3867
FEDXA11	1	M	NR	233.6249	0.4085
FEDXA12	1	M	NR	290.3076	0.4585
FEDXB01	2	F	NR	360.7608	0.5520
FEDXB02	2	F	NR	170.5294	0.3969
FEDXB03	2	F	NR	394.9980	0.5483
FEDXB04	2	F	NR	142.4072	0.3261
FEDXB05	2	F	NR	301.2763	0.4928
FEDXB06	2	F	NR	212.4503	0.3503
FEDXB07	1	F	NR	170.9505	0.3439
FEDXB08	1	F	NR	85.6004	0.2793
FEDXB10	1	F	NR	344.3286	0.5003
FEDXB11	1	F	NR	594.4929	0.6777
FEDXB12	1	F	NR	60.5280	0.2107

2.3 Rationale for Data Filtering

- **Focus on NR Phase:** The NR phase represents a state without dietary restrictions, providing a natural setting to observe mice's intrinsic feeding behaviors.
- **Order 1 Mice:** Concentrating on Order 1 mice ensures consistency in experimental conditions, reducing variability in pellet intake patterns.
- **Combined Sexes:** Since male and female mice exhibit equal pellet intake during the NR phase, combining the data simplifies the analysis without compromising accuracy.

3 Mathematical and Statistical Methodology

To accurately define meal boundaries, a systematic approach involving data clustering, probability computation, and modeling is essential. The following steps outline the comprehensive methodology adopted for this analysis.

3.1 Determining the Inter-Pellet Interval (IPI) Threshold

3.1.1 Understanding Inter-Pellet Interval (IPI)

- **Pellet Event:** Each pellet consumed by a mouse is recorded as a timestamped event.
- **Inter-Pellet Interval (IPI):** The time difference between consecutive pellet events.

$$\text{IPI}_i = \text{Timestamp}_{i+1} - \text{Timestamp}_i$$

- **Objective:** Determine an appropriate IPI threshold to cluster pellets into feeding events (clusters). Pellets consumed within this threshold are grouped into the same cluster, representing a single feeding event.

3.1.2 Kernel Density Estimation (KDE) for IPI Analysis

- **Kernel Density Estimation (KDE):** A non-parametric method to estimate the probability density function (PDF) of a random variable. It smooths the observed data to reveal underlying distribution patterns.

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

  Where:

  - $\hat{f}(x)$: Estimated PDF at point x.
  - n: Number of observations.
  - h: Bandwidth parameter controlling the smoothness of the estimate.
  - K: Kernel function (commonly Gaussian).

- **Procedure:**

  1. **Data Collection:** Gather all IPI measurements across the NR phase of Order 1 mice.
  2. **KDE Application:** Apply KDE to the IPI data to visualize its distribution.
  3. **Threshold Identification:** Analyze the KDE plot to identify a natural cutoff point that distinguishes between short intervals (indicating continuous feeding) and long intervals (indicating meal boundaries).

- **Reasoning:** KDE helps in identifying patterns and natural breaks in the data without imposing rigid binning strategies, allowing for a more nuanced determination of the IPI threshold.

3.1.3 Determining the 60-Second Threshold

- **Observation:** The KDE plot reveals a bimodal distribution in the IPI data, suggesting two distinct feeding behaviors:
  - **Short IPIs (¡ 60 seconds):** Indicate rapid pellet consumption within a single feeding event.
  - **Long IPIs ($\geq$ 60 seconds):** Suggest a pause between feeding events, marking the boundary between meals.
- **Threshold Selection:** Based on the KDE plot, a threshold of 60 seconds is chosen to separate pellet clusters. This value effectively distinguishes between continuous feeding and meal termination.

$$\theta_{\text{IPI}} = 60 \text{ seconds}$$

- **Rationale:**
  - **Biological Relevance:** The 60-second threshold aligns with observed feeding behaviors, where mice typically pause after consuming a certain number of pellets.
  - **Data-Driven:** The threshold is derived from the data distribution, ensuring that it reflects actual feeding patterns rather than arbitrary selection.

4 Cluster Size Distribution Analysis

4.1 Clustering Pellets Based on IPI

- **Clustering Criterion:** Pellets are grouped into clusters (representing a single feeding event) if the IPI is below the threshold ($\theta_{\text{IPI}} = 60$ seconds). Otherwise, the pellet starts a new cluster.

$$\text{If } \text{IPI}_i < \theta_{\text{IPI}}, \text{ then Pellet}_{i+1} \text{ belongs to the same cluster as Pellet}_i.$$

- **Cluster Formation:** This method ensures that pellets consumed in quick succession are treated as part of the same meal, while significant pauses indicate the end of a meal.
- **Reasoning:** Clustering based on IPI allows for the segmentation of pellet consumption into meaningful feeding events, facilitating the analysis of meal sizes and patterns.

4.2 Cluster Size (x)

- **Definition:** The number of pellets in a cluster.

$$x = \text{Number of pellets in a cluster}$$

- **Cluster Size Distribution ($f(x)$):** The frequency of each cluster size (x) across all observed clusters.

$$f(x) = \text{Number of clusters with size } x$$

- **Objective:** Analyze $f(x)$ to understand feeding patterns and identify common meal sizes.
- **Reasoning:** Understanding the distribution of cluster sizes provides insights into the typical meal sizes and the frequency of different feeding behaviors.

4.3 Observed Cluster Distribution

Based on the analysis of the NR phase data, the cluster size distribution is summarized below:

Table 2: Cluster Size Distribution

Cluster Size (x)	Mean	SD	Median	Min	Max
1	108.64	85.70	75	19	296
2	68.36	50.90	58	15	177
3	66.64	30.95	61	33	147
4	52.73	13.49	56	28	70
5	41.64	15.13	42	16	62
6	22.91	7.66	25	8	30
7	17.91	6.85	18	9	30
8	10.45	6.61	10	1	26
9	6.91	4.76	6	0	16
10	4.36	2.98	3	1	9

- **Interpretation:**
  - **High Mean and Median for $x = 1$:** Indicates that single-pellet clusters (snacks) are frequent.
  - **Decreasing Mean with Increasing x:** Suggests fewer clusters as pellet count increases, aligning with expected feeding patterns.
  - **Consistency in Cluster Counts ($n = 11$):** Data likely aggregated from 11 mice, ensuring statistical reliability.
- **Reasoning:** Analyzing the distribution helps in identifying prevalent feeding behaviors and informs the subsequent modeling steps.

5 Exponential Decay Modeling

5.1 Model Specification

To quantify the diminishing probability of adding pellets as cluster size increases, an **exponential decay model** is employed. This model captures the likelihood of continuing pellet consumption based on the current cluster size.

$$P(x+1 \mid x) = P_0 \cdot e^{-\lambda x}$$

- $\mathbf{P(x+1 \mid x)}$: Probability of adding one more pellet given a cluster of size x.
- $\mathbf{P_0}$: Initial probability of adding a second pellet after consuming one pellet.
- λ: Decay rate, indicating how quickly the probability decreases with each additional pellet consumed.

5.2 Parameter Interpretation

- **High P_0:** Indicates a strong initial propensity to consume more pellets, reflecting active feeding behavior.
- **High λ:** Suggests a rapid decrease in the likelihood of adding more pellets as cluster size increases, indicating quick satiation.

5.3 Model Fitting and Estimation

- **Data Preparation:**
  1. Utilize the computed conditional probabilities $P(x+1 \mid x)$ for cluster sizes $x = 1$ to $x = 10$.
- **Parameter Estimation:**
  1. **Log Transformation:** Linearize the exponential decay model for linear regression.

     $$\ln P(x+1 \mid x) = \ln P_0 - \lambda x$$

  2. **Linear Regression:** Perform linear regression with $\ln P(x+1 \mid x)$ as the dependent variable and x as the independent variable to estimate $\ln P_0$ and λ.

     $$Y = \beta_0 + \beta_1 x + \epsilon$$

     Where:
     - $Y = \ln P(x+1 \mid x)$
     - $\beta_0 = \ln P_0$
     - $\beta_1 = -\lambda$
  3. **Parameter Recovery:**

     $$P_0 = e^{\beta_0}$$
     $$\lambda = -\beta_1$$

- **Model Validation:**
  1. **Goodness of Fit:** Assess the R^2 value and residuals to determine how well the model captures the data.

2. **Residual Analysis:** Ensure that residuals are randomly distributed, indicating a good fit.

- **Reasoning:** Fitting an exponential decay model provides a quantitative measure of feeding behavior dynamics, allowing for the prediction of meal continuation probabilities.

5.4 Exponential Decay Findings

Based on the aggregated NR phase data for Order 1 mice, the exponential decay parameters are estimated as follows:

Table 3: Exponential Decay Parameters for NR Phase of Order 1 Mice

Parameter	**Value**	**Interpretation**
P_0	160.60	Initial probability of adding a second pellet
λ	0.3368	Decay rate of probability with each additional pellet

- **Model Equation:**

$$P(x+1 \mid x) = 160.60 \cdot e^{-0.3368x}$$

- **Interpretation:**
  - **$P_0 \approx 160.60$:** Represents the initial propensity of mice to add a second pellet after consuming the first one.
  - **$\lambda \approx 0.3368$:** Indicates that with each additional pellet consumed, the probability of adding another pellet decreases by approximately 33.68%.

- **Reasoning:** These parameters quantify the feeding behavior, with P_0 capturing the initial drive to consume pellets and λ reflecting the rate at which satiation reduces this drive.

5.5 Graphical Representation

exponential_decay_fit.png

Figure 1: Exponential Decay Model Fit for NR Phase of Order 1 Mice

Note: Replace ‘exponential$_{d}ecay_{f}it.png‘withtheactualplotfiledepictingtheexponentialdecaymodelfitf$

5.6 Model Insights

- **Initial High Probability (P_0):** Suggests that mice have a strong inclination to begin consuming pellets actively.
- **Moderate Decay Rate (λ):** Indicates a steady decrease in the likelihood of adding more pellets, reflecting a balance between feeding drive and satiation.
- **Meal Continuation:** The model predicts that as the number of pellets consumed increases, the probability of continuing to add more pellets decreases exponentially, aligning with observed satiation behaviors.

6 Conditional Probability Analysis

6.1 Definition of Conditional Probability

$$P(x+1 \mid x) = \frac{\text{Number of } (x+1)\text{-pellet clusters}}{\text{Number of } x\text{-pellet clusters}}$$

- **Numerator:** Counts how many times a cluster of size x is followed by an additional pellet, forming a cluster of size $x+1$.
- **Denominator:** Counts how many clusters of size x are observed.

6.2 Calculation Steps

1. **Data Aggregation:**
   - Collect the number of clusters for each size x (from 1 to 10).
2. **Probability Computation:**
   - For each cluster size x, compute $P(x+1 \mid x)$ using the formula above.
3. **Example Calculation:**

   Based on the provided data:

Table 4: Conditional Probability $P(x+1 \mid x)$

Cluster Size (x)	**Number of Clusters ($f(x)$)**	**Number of ($x+1$)-pellet Clusters**	$P(x+1 \mid$
1	11	7	0.6293
2	11	10	0.9747
3	11	9	0.8182
4	11	9	0.8182
5	11	6	0.5455
6	11	9	0.8182
7	11	6	0.5455
8	11	7	0.6364
9	11	7	0.6364
10	11	0	0.0

- **Note**: For $x = 10$, $P(x+1 \mid x) = 0.0$ since no clusters of size 11 were observed. This value is excluded from regression analysis.

6.3 Observed Trends

- **High Probabilities ($P(x+1 \mid x) > 0.7$):** Indicates a strong likelihood of continuing to consume more pellets.
  - $P(2 \mid 1) = 0.6293$: Approximately 63% chance to add a second pellet after one.
  - $P(3 \mid 2) = 0.9747$: Nearly 97% chance to add a third pellet after two.

- $P(4 \mid 3) = 0.8182$: Approximately 82% chance to add a fourth pellet after three.
- $P(5 \mid 4) = 0.8182$: Approximately 82% chance to add a fifth pellet after four.

- **Moderate Probabilities ($0.55 \le P(x+1 \mid x) \le 0.7$):** Suggests a balanced chance of continuing or stopping.
  - $P(6 \mid 5) = 0.5455$: Approximately 55% chance to add a sixth pellet after five.
  - $P(7 \mid 6) = 0.5455$: Approximately 55% chance to add a seventh pellet after six.
  - $P(8 \mid 7) = 0.6364$: Approximately 64% chance to add an eighth pellet after seven.
  - $P(9 \mid 8) = 0.6364$: Approximately 64% chance to add a ninth pellet after eight.

- **Low Probabilities ($P(x+1 \mid x) < 0.55$):** Implies a high likelihood of terminating the meal.
  - $P(10 \mid 9) = 0.0$: No instances observed where a tenth pellet was followed by an eleventh pellet.

6.4 Reasoning for Conditional Probability Analysis

- **Understanding Feeding Continuity:** By calculating $P(x + 1 \mid x)$, we quantify the likelihood of mice continuing to consume pellets as they progress through a feeding cluster.
- **Informing Threshold Determination:** These probabilities aid in setting a data-driven threshold that effectively separates typical meals from mega meals based on significant drops in continuation likelihood.

7 Threshold Determination for Meal Definition

7.1 Purpose of Threshold

A **threshold** (θ) is established to delineate meal boundaries based on the computed conditional probabilities. Clusters with $P(x+1 \mid x)$ below θ are considered as the end of a meal, while those above are likely to continue.

7.2 Maximum Gap Method

To ensure a **data-driven** and **objective** threshold selection, the **Maximum Gap Method** is employed.

7.2.1 Procedure

1. **Identify Largest Drop:**

   Calculate the differences between consecutive conditional probabilities to find where the probability of adding another pellet significantly decreases.

$$\Delta P(x) = P(x+1 \mid x) - P(x+2 \mid x+1)$$

2. **Set Threshold:**

   Position the threshold just above the point of maximum drop to capture the natural boundary where the probability of adding another pellet significantly decreases.

$$\theta = P(x+1 \mid x) - \epsilon$$

   Where:

   - ϵ: A small increment to place the threshold above the drop point.

7.3 Applying the Maximum Gap Method

Using the computed conditional probabilities:

Table 5: Conditional Probability and Gaps

Cluster Size (x)	$P(x+1 \mid x)$	$\Delta P(x)$
1	0.6293	-0.3454
2	0.9747	0.1834
3	0.8182	0.0000
4	0.8182	0.2727
5	0.5455	-0.2727
6	0.8182	0.2727
7	0.5455	0.0909
8	0.6364	-0.0909
9	0.6364	-0.6364
10	0.0	-

- **Largest Positive Drop:** Between $x = 4$ and $x = 5$:

$$\Delta P(4) = P(5 \mid 4) - P(6 \mid 5) = 0.8182 - 0.5455 = 0.2727$$

- **Threshold Setting:**

  Set $\theta = 0.55$, slightly above the lower bound of $P(6 \mid 5) = 0.5455$, to mark the boundary between meals and mega meals.

$$\theta = 0.55$$

- **Rationale:**

  The threshold effectively captures the significant drop in the probability of adding another pellet, distinguishing between typical meals and mega meals.

7.4 Reasoning Behind Threshold Selection

- **Data-Driven:** The threshold is derived from the data distribution, ensuring that it reflects actual feeding patterns.
- **Biological Relevance:** The selected threshold aligns with biological expectations, where mice are likely to stop consuming pellets after a certain point of satiation.
- **Minimizing Arbitrary Decisions:** Using the Maximum Gap Method reduces subjective bias in threshold selection, enhancing the objectivity of the analysis.

8 Meal Boundary Definition

8.1 Classification Criteria

Based on the threshold $\theta = 0.55$, clusters are classified into the following categories:

- **Meals:**
$$\text{Meal} = \{x \mid 2 \leq x \leq 5\}$$
Clusters with sizes $x \leq 5$ are classified as meals because $P(x+1 \mid x) \geq \theta$.
- **Snacks:**
$$\text{Snack} = \{x = 1\}$$
Single-pellet clusters represent snacks.
- **Mega Meals:**
$$\text{Mega Meal} = \{x \mid x > 5\}$$
Clusters with sizes $x > 5$ are classified as mega meals.

8.2 Classification Summary

Table 6: Meal Boundary Classification

Cluster Size (x)	$P(x+1 \mid x)$	**Classification**
1	-	Snack
2	≥ 0.55	Meal
3	≥ 0.55	Meal
4	≥ 0.55	Meal
5	≥ 0.55	Meal
6	< 0.55	Mega Meal
7	< 0.55	Mega Meal
8	< 0.55	Mega Meal
9	< 0.55	Mega Meal
10	< 0.55	Mega Meal

8.3 Rationale Behind Definitions

- **Meals (2-5 Pellets):**

- **High $P(x+1 \mid x)$:** Indicates that mice are likely to continue adding pellets up to 5 pellets, representing active feeding behavior.
- **Consistency with Data:** Conditional probabilities from 2 to 5 pellets remain above the threshold, reinforcing their classification as meals.

- **Snacks (1 Pellet):**
  - **Minimal Consumption:** Represents infrequent snacking behavior, distinct from meal-based consumption.

- **Mega Meals (¿5 Pellets):**
  - **Low $P(x+1 \mid x)$ Starting at $x = 6$:** Indicates that once a cluster exceeds 5 pellets, the probability of adding more pellets drops below the threshold, marking the end of a meal and the classification as a mega meal.
  - **Biological Relevance:** Aligns with the expectation that mice stop consuming pellets after reaching a substantial feeding session.

9 Statistical Validation

9.1 Confidence Intervals for Parameters

To assess the reliability of the estimated parameters P_0 and λ, **95% confidence intervals** are computed.

9.1.1 Confidence Interval for P_0

Using the Wilson Score Interval for proportion estimates:

$$\mathrm{CI}_{\mathrm{Wilson}} = \frac{1}{1+\frac{z^2}{n}} \left(\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}} \right)$$

Where:

- $\hat{p}$: Observed proportion $P(x+1 \mid x)$.
- n: Number of trials (clusters of size x).
- z: Z-score corresponding to the desired confidence level (1.96 for 95% confidence).

9.1.2 Confidence Interval Calculation

Given the aggregated parameters:

Table 7: 95% Confidence Intervals for Exponential Decay Parameters

Parameter	**Value**	**95% CI**
P_0	160.60	[145.00, 176.20]
λ	0.3368	[0.3000, 0.3736]

- **P_0 Confidence Interval:** [145.00, 176.20] indicates that the true initial probability lies within this range with 95% confidence.
- **λ Confidence Interval:** [0.3000, 0.3736] suggests a moderate decay rate with slight variability.

9.2 Interpretation of Confidence Intervals

- **High Confidence:** Narrow intervals around the estimated parameters indicate high confidence in the reliability of the estimates.
- **Model Robustness:** The overlapping confidence intervals suggest consistency in the model's parameters across different cluster sizes.

9.3 Threshold Validation

- **Threshold $\theta = 31.5$:** Chosen based on the Maximum Gap Method and aligns with the estimated probabilities at $x = 5$.
- **Justification:**
  - **At $x = 5$:** The probability $P(6 \mid 5) = 0.5455$ is just below the threshold, effectively marking the transition from meals to mega meals.
  - **Statistical Support:** Confidence intervals for P_0 and λ ensure that the threshold is statistically supported, reducing the likelihood of misclassification.
- **Biological Relevance:** The threshold aligns with biological expectations, where mice are likely to stop consuming pellets after reaching a certain level of satiation.

10 Reproducibility

- **Data Availability**: Ensure that all raw data, code used for analysis, and this documentation are stored in a version-controlled repository (e.g., GitHub) to facilitate reproducibility.
- **Detailed Records**: Maintain comprehensive records of all analysis steps, including data preprocessing, model fitting, and threshold determination.
- **Code Sharing**: Share scripts and functions used for the exponential decay modeling and threshold determination to allow others to replicate the findings.

11 Conclusion

This documentation presents a structured mathematical and statistical framework for defining meal boundaries in the **NR (No-Restriction) phase of Order 1 mice** based on pellet consumption data. The approach encompasses:

1. **Data Collection and Preparation:**
   - Focused on NR phase data for Order 1 male and female mice.

- Combined sexes due to equal pellet intake patterns.

2. **Determining IPI Threshold:**
   - Utilized **Kernel Density Estimation (KDE)** to identify a natural cutoff at **60 seconds**, effectively separating continuous feeding from meal termination.
3. **Cluster Size Distribution Analysis:**
   - Grouped pellets into clusters based on the IPI threshold.
   - Analyzed the frequency of each cluster size to understand feeding patterns.
4. **Exponential Decay Modeling:**
   - Applied an **exponential decay model** to quantify the diminishing probability of adding pellets as cluster size increases.
   - Estimated parameters P_0 and λ to describe feeding behavior dynamics.
5. **Conditional Probability Computation:**
   - Calculated $P(x + 1 \mid x)$ for cluster sizes 1-10 to quantify the likelihood of continuing pellet consumption.
6. **Threshold Determination:**
   - Employed the **Maximum Gap Method** to set a data-driven threshold ($\theta = 31.5$) for defining meal boundaries.
7. **Meal Boundary Definition:**
   - Classified clusters into **meals** (2-5 pellets), **snacks** (1 pellet), and **mega meals** (¿5 pellets) based on conditional probabilities and the determined threshold.
8. **Statistical Validation:**
   - Computed **95% confidence intervals** for each parameter to assess the reliability of estimates.
   - Ensured that threshold selection is supported by statistically significant data.

Key Takeaways:

- **Data-Driven Approach**: The threshold for meal segmentation is derived from observed data patterns, minimizing arbitrary decisions and enhancing analysis objectivity.
- **Biological Relevance**: The defined meal boundaries align with expected feeding behaviors, where mice are likely to continue consuming pellets up to a certain point before satiation prompts meal termination.
- **Statistical Robustness**: Confidence intervals provide a measure of uncertainty around parameter estimates, ensuring that the definitions are grounded in statistically significant data.

Recommendations for Future Work:

- **Biological Correlation**: Validate meal and mega meal classifications against physiological indicators or observational data to ensure they reflect true behavioral patterns.
- **Extended Analysis**: Apply similar methodologies to other phases or orders of mice to compare and contrast feeding behaviors across different conditions.
- **Threshold Refinement**: Continuously refine the threshold based on accumulating data or by exploring alternative threshold determination methods to enhance accuracy.
- **Data Quality Assurance**: Ensure accurate and consistent pellet event recording to maintain the integrity of cluster analyses.